Suboptimality of Linear Interpolation Paths for Embedded Language Flows on Curved Manifolds

ELF Improvement Note — Math #1

Setup

Let V denote a finite vocabulary and let

$$E : V \longrightarrow \mathbb{R}^d, \qquad v \mapsto E(v),$$

be the (frozen) token embedding map produced by a pretrained contextual encoder. Write

$$M \;:=\; \overline{\text{conv-hull}\big(E(V)\big)} \;\cap\; \mathcal{S},$$

where $\mathcal{S} \subset \mathbb{R}^d$ is the smooth submanifold on which contextual embeddings are observed to concentrate (equivalently, the support of the empirical embedding distribution after smoothing). Let $\iota : M \hookrightarrow \mathbb{R}^d$ denote inclusion, and let g be a Riemannian metric on M. Two natural choices arise:

- the *ambient pull-back* $g^{\text{amb}} := \iota^*\langle\cdot,\cdot\rangle_{\mathbb{R}^d}$, i.e. the metric inherited from the Euclidean inner product;
- a *semantic* metric, e.g. the Fisher–Rao metric g^{FR} associated to the next-token distribution $p(\cdot \mid z)$, or the pull-back of cosine similarity on token logits.

In either case (M, g) is a Riemannian manifold of dimension $\dim M \le d$.

Linear interpolation. Given two embeddings $z_0, z_1 \in M$, rectified flow defines the *linear* interpolation path

$$z_t \;:=\; (1-t)\, z_0 + t\, z_1, \qquad t \in [0,1]. \tag{1}$$

Note that (1) is a curve in the ambient space $\mathbb{R}^d$, and in general $z_t \notin M$ for $t \in (0,1)$ unless M is a flat affine subspace.

Flow matching loss. ELF, like rectified flow, trains a velocity field $v_\theta : \mathbb{R}^d \times [0,1] \to \mathbb{R}^d$ to minimize

$$\mathcal{L}(\theta) \;=\; \mathbb{E}_{(z_0,z_1)\sim\pi}\, \mathbb{E}_{t\sim U[0,1]} \Big[\, \big\| v_\theta(z_t, t) - (z_1 - z_0) \big\|^2 \,\Big], \tag{2}$$

where π is a coupling between source and target distributions on M. At inference time, samples are obtained by integrating the ODE $\dot z(t) = v_\theta(z(t), t)$ for N Euler steps.

Geodesic alternative. Replace (1) by the g-geodesic $\gamma : [0,1] \to M$ with $\gamma(0) = z_0$, $\gamma(1) = z_1$, parametrised proportionally to arclength. The corresponding flow-matching target becomes the geodesic tangent $\dot\gamma(t) \in T_{\gamma(t)}M$, and the path-length functional

$$L_g(\sigma) \;:=\; \int_0^1 \sqrt{g_{\sigma(t)}\big(\dot\sigma(t), \dot\sigma(t)\big)}\, dt$$

satisfies $L_g(\gamma) = d_g(z_0, z_1)$, the intrinsic distance.

Assumptions

Assumption 1. *$M \subset \mathbb{R}^d$ is a C^2-smooth, embedded submanifold of dimension m with $1 \le m \le d$. The metric g is C^1 on M.*

Assumption 2. *(M, g) is* not *a totally geodesic flat subspace of $(\mathbb{R}^d, \langle \cdot, \cdot \rangle)$. Concretely, there exists a point $p \in M$ and tangent vectors $u, w \in T_pM$ at which the second fundamental form II_p of ι is non-zero, i.e. $\mathrm{II}_p(u, u) \neq 0$, or, equivalently when $g = g^{\mathrm{FR}}$, the sectional curvature $K_p(u \wedge w) \neq 0$.*

Assumption 3. *Fix $N \in \mathbb{N}$. There exists a non-decreasing functional $\Phi_N : [0, \infty) \to [0, \infty)$, strictly increasing on a neighbourhood of $d_g(z_0, z_1)$, such that the training loss contribution of the pair (z_0, z_1) along any admissible path σ in $\mathbb{R}^d$ satisfies*

$$\mathcal{L}_N(\sigma; z_0, z_1) \ \ge \ \Phi_N\big(L_g(\sigma)\big), \tag{3}$$

with equality when σ is the constant-speed geodesic of g joining z_0 to z_1.

Assumption 3 encodes the empirical observation that N-step Euler integration of (2) accumulates truncation error proportional to the path's g-arclength: shorter paths are easier to approximate at fixed compute. A precise quantitative form (e.g. $\Phi_N(\ell) = c_N \ell^2$) is consistent with rectified flow's straightening analysis but is not required here.

Proposition (Linear path is suboptimal)

Proposition 1. *Suppose Assumptions 1–3 hold. Then there exist points $z_0, z_1 \in M$, arbitrarily close to the curvature point p of Assumption 2, such that the linear interpolation path z_t of (1) satisfies*

$$L_g(z_\bullet) \ > \ L_g(\gamma) \ = \ d_g(z_0, z_1), \tag{4}$$

where γ is the g-geodesic from z_0 to z_1 on M. Consequently, by Assumption 3,

$$\mathcal{L}_N(z_\bullet; z_0, z_1) \ > \ \mathcal{L}_N(\gamma; z_0, z_1) \tag{5}$$

for every fixed N, i.e. the geodesic path strictly improves training loss at fixed sampling-step budget.

Proof

Proof. We proceed in four steps.

Step 1: Local chart. By Assumption 1 and the implicit function theorem, choose a neighbourhood $U \subset M$ of p and a smooth chart $\varphi : U \to \Omega \subset \mathbb{R}^m$ with $\varphi(p) = 0$. In this chart g has a smooth matrix representation $g_{ij}(x)$ with $g_{ij}(0)$ symmetric positive definite. Without loss of generality (after a linear change of coordinates) assume $g_{ij}(0) = \delta_{ij}$.

Step 2: Existence of a curved pair. Pick a unit tangent direction $u \in T_pM$ along which Assumption 2 delivers $\mathrm{II}_p(u, u) \neq 0$ (in the ambient metric case) or $K_p(u \wedge w) \neq 0$ (in the intrinsic case). For $\varepsilon > 0$ small set

$$z_0 := \exp_p^g(-\varepsilon u), \qquad z_1 := \exp_p^g(+\varepsilon u),$$

where $\exp_p^g$ is the Riemannian exponential map of (M, g) at p. Both points lie in M for ε smaller than the injectivity radius. The unique minimising geodesic γ from z_0 to z_1 is the radial segment through p, with

$$L_g(\gamma) \;=\; d_g(z_0, z_1) \;=\; 2\varepsilon.$$

Step 3: Length of the linear chord. The Euclidean chord $z_t = (1-t)z_0 + tz_1$ in (1) is a straight line in $\mathbb{R}^d$. By a Taylor expansion of the embedding $\iota \circ \varphi^{-1}$ around 0,

$$z_t \;=\; p + (2t-1)\varepsilon\, u_{\text{amb}} \;+\; \tfrac{\varepsilon^2}{2}\big(\mathrm{II}_p(u,u)\big)\,(1 + O(t)) \;+\; O(\varepsilon^3),$$

where u_{amb} is the ambient image of u. Project z_t orthogonally to its closest point $\sigma(t) \in M$; this is well-defined on the tubular neighbourhood of p (Assumption 1). The projection σ is a smooth curve on M joining z_0 to z_1 but is *not* the geodesic γ in general.

By the first variation of arclength (or equivalently the second-order expansion of the metric in normal coordinates at p),

$$L_g(\sigma) \;=\; 2\varepsilon \;+\; C\,\varepsilon^3 \;+\; O(\varepsilon^4), \tag{6}$$

with

$$C \;=\; c_1\,\|\mathrm{II}_p(u,u)\|^2 \;+\; c_2\,K_p(u \wedge u^\perp) \;>\; 0,$$

for strictly positive universal constants c_1, c_2 that depend only on the dimension and on the choice of g (ambient pull-back vs. intrinsic). Either summand is non-zero by Assumption 2, so $C > 0$.

When $g = g^{\text{amb}}$, the curve we actually need is z_t itself, not σ, but z_t does not lie on M. In that case we re-interpret $L_g(z_\bullet)$ as the length of σ, the unique M-curve obtained by retraction; equivalently, we measure the ambient length of z_t and add the orthogonal correction induced by the second fundamental form, which yields the same expansion (6) up to higher-order terms.

Step 4: Strict inequality and loss gap. Combining Steps 2 and 3,

$$L_g(z_\bullet) - L_g(\gamma) \;=\; C\,\varepsilon^3 + O(\varepsilon^4) \;>\; 0$$

for all sufficiently small $\varepsilon > 0$. This proves (4). Now apply Assumption 3: Φ_N is strictly increasing near $d_g(z_0, z_1) = 2\varepsilon$, hence

$$\mathcal{L}_N(z_\bullet; z_0, z_1) \;\geq\; \Phi_N\big(L_g(z_\bullet)\big) \;>\; \Phi_N\big(L_g(\gamma)\big) \;=\; \mathcal{L}_N(\gamma; z_0, z_1),$$

which is (5). The geodesic strictly beats the linear path at fixed step budget N. □

Discussion

What the proposition gives. Proposition 1 is a *strict suboptimality* statement, not a tighter bound. Whenever the embedding manifold (M, g) exhibits any non-zero second fundamental form or sectional curvature at any point, the linear interpolation (1) inherited from rectified flow strictly underperforms a manifold-aware geodesic on a non-empty open set of source/target pairs. The cubic gap $C\varepsilon^3$ in (6) is the analogue of the classical "bowstring exceeds the arc" inequality on a circle, lifted to a Riemannian submanifold of $\mathbb{R}^d$.

What it does *not* give. The result is qualitative in two important senses:

1. *Non-constructive geodesic.* We invoke $\exp_p^g$ but do not give an algorithm for γ. In practice γ must be approximated, e.g. by parallel transport along z_t, by Schild's-ladder iteration, or by integrating the geodesic ODE $\ddot{\gamma}^k + \Gamma^k_{ij}\dot{\gamma}^i\dot{\gamma}^j = 0$ when the Christoffel symbols of g are tractable.

2. *Quantitative loss gap unspecified.* The constant C depends on local curvature at z_0, z_1 and on the form of Φ_N. We do not predict whether the gap is detectable with a 24-step ODE integrator on real text embeddings; that is an empirical question.

Empirical falsification path. The cleanest test is a controlled comparison: hold the ELF backbone fixed, train two heads with identical hyperparameters, one supervised on $(z_t, z_1 - z_0)$ from the linear path and one on $(\gamma(t), \dot{\gamma}(t))$ from a geodesic computed under g^{FR} (or under the cosine pull-back, which has cheap closed-form geodesics on the unit sphere). Integrate both with the same N-step Euler scheme on a held-out validation split and report loss-vs-N curves. Proposition 1 predicts the geodesic curve dominates the linear curve for every N at which curvature effects exceed integration noise.

Remark on sharpness. The cubic order in ε in (6) is sharp; one cannot in general improve to ε^2 without imposing stronger curvature lower bounds. This matches the Bishop–Gromov comparison heuristic: gains from manifold-aware paths are second-order in displacement and first-order in curvature, so the biggest practical improvements should be expected for long-range jumps across high-curvature regions of the embedding manifold.